

KAUNAS INTL, Lithuania

WMO: 266295

Lat: 54.964N

Lon: 24.085E

Elev: 78

StdP: 100.39

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.9	-15.2	-21.8	0.5	-17.8	-18.9	0.7	-14.9	11.6	3.7	10.1	3.1	1.9	80	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	29.1	20.0	27.1	19.3	25.2	18.2	21.5	26.6	20.4	25.3	19.4	23.8	3.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	14.8	23.9	18.8	13.8	22.6	17.8	12.9	21.7	63.0	26.8	58.9	25.2	55.4	24.4	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.6	6.9	-21.3	31.9	3.7	2.2	-24.0	33.5	-26.2	34.8	-28.3	36.1	-31.0	37.7		
(5)			-21.6	23.2	3.6	0.9	-24.2	23.9	-26.3	24.4	-28.4	24.9	-31.0	25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-4.0	-2.9	1.5	7.7	13.0	16.3	18.7	18.0	13.5	7.6	3.6	-0.1
	DBStd	8.85	6.25	5.72	4.55	4.07	4.07	3.37	2.93	3.04	3.12	3.72	3.71	4.61
	HDD10.0	1772	434	361	265	90	14	1	0	0	7	92	193	313
	HDD18.3	3949	693	594	522	318	172	80	31	44	148	332	443	572
	CDD10.0	967	0	0	1	23	108	188	269	248	111	19	0	0
	CDD18.3	103	0	0	0	0	7	18	42	34	2	0	0	0
	CDH23.3	932	0	0	0	7	81	191	351	282	19	0	0	0
	CDH26.7	222	0	0	0	1	12	41	90	76	3	0	0	0
Wind	WSAvg	3.4	3.9	3.7	3.7	3.4	3.2	3.1	2.9	2.8	3.0	3.4	3.7	3.9
Precipitation	PrecAvg	688	50	41	39	39	61	72	91	75	57	59	52	51
	PrecMax	877	99	78	90	80	102	170	205	166	126	157	101	76
	PrecMin	489	15	13	12	8	29	13	7	6	18	9	11	20
	PrecStd	84	20	14	19	18	20	29	49	40	29	30	20	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.9	7.9	15.2	24.1	28.1	30.2	31.9	32.0	26.0	18.8	12.1	9.2
		MCWB	7.9	5.9	8.3	14.3	18.4	19.9	20.8	21.6	18.5	14.1	9.8	8.2
	2%	DB	6.1	5.9	12.1	20.9	25.8	27.8	29.1	28.8	22.9	16.2	10.2	8.0
		MCWB	5.1	4.1	6.6	12.0	17.0	19.0	20.1	20.5	16.8	13.3	9.0	6.9
	5%	DB	4.9	4.2	9.8	18.1	23.2	25.2	27.0	26.2	20.9	14.2	9.2	6.2
		MCWB	4.0	2.9	6.2	10.1	15.4	17.4	19.8	18.9	16.1	11.8	8.3	5.4
	10%	DB	3.2	3.1	7.2	15.8	21.1	23.2	25.1	24.1	19.0	13.0	8.1	5.2
		MCWB	2.5	2.1	4.9	9.2	14.6	16.5	18.8	18.3	14.7	11.1	7.4	4.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	5.7	9.6	15.3	19.8	21.4	23.4	23.5	19.5	15.3	10.7	8.4
		MCDB	8.7	7.3	13.6	23.2	26.2	27.4	29.1	29.1	24.5	17.9	11.4	8.9
	2%	WB	5.1	4.2	7.8	12.8	18.0	19.9	21.8	21.2	17.7	13.5	9.4	6.9
		MCDB	5.7	5.2	10.6	18.8	23.6	25.6	26.6	26.6	21.7	15.7	10.0	7.4
	5%	WB	3.7	3.0	6.4	11.1	16.8	18.6	20.7	20.1	16.6	12.4	8.3	5.4
		MCDB	4.4	3.9	8.8	16.5	22.1	23.8	25.5	24.4	19.7	14.3	9.0	6.2
	10%	WB	2.5	2.1	4.9	9.8	15.3	17.5	19.6	18.8	15.5	11.4	7.4	4.2
		MCDB	3.1	2.9	6.7	14.4	19.8	22.2	23.4	23.3	18.4	13.0	7.9	4.9
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.0	7.1	10.5	10.9	10.4	9.7	9.8	8.8	6.6	3.8	3.5
		MCDBR	4.3	5.0	10.7	15.0	13.8	13.5	12.8	13.3	11.7	8.2	4.8	3.9
		MCWBR	3.9	4.0	6.7	8.0	7.0	6.5	5.8	6.2	6.8	5.6	4.1	3.7
	5% WB	MCDBR	3.7	4.4	8.8	12.8	12.4	11.5	11.0	11.4	10.2	7.4	4.4	3.9
		MCWBR	3.6	3.8	6.1	7.6	7.0	6.2	5.6	6.0	6.4	5.3	4.0	3.8
Clear-Sky Solar Irradiance	taub	0.298	0.331	0.367	0.409	0.396	0.388	0.424	0.419	0.382	0.357	0.338	0.300	
	taud	2.196	2.180	2.211	2.193	2.292	2.343	2.262	2.273	2.349	2.396	2.279	2.231	
	Ebn at Noon	617	728	793	810	847	858	815	793	775	705	566	529	
	Edh at Noon	60	88	108	127	121	117	124	116	94	72	57	48	
All-Sky Solar Radiation	RadAvg	0.57	1.32	2.60	4.06	5.31	5.53	5.16	4.33	2.98	1.46	0.53	0.36	
	RadStd	0.12	0.25	0.41	0.52	0.51	0.47	0.50	0.41	0.33	0.28	0.07	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)		+0.41	N/A	N/A	N/A	N/A	+0.55	N/A	N/A	N/A	N/A

Nomenclature: See separate page